Exercise 3 CASE STATEMENTS

Questions

1. Table products

Product_id	product_name	Price
1	laptop	1200,00
2	Phone	800.00
3	keyboard	45.00
4	Monitor	300,00
5	Mouse	25.00

SQL

```
SELECT product_name,
       price,
CASE
        WHEN price > 1000 THEN 'Expensive'
        WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
        ELSE 'Budget'
END AS price_category
FROM products;
```

Product	Price	Price_category
Laptop	1200	Expensive
Phone	800	Mid-Range
Keyboard	45	Budget
Monitor	300	Mid-Range
Mouse	25	Budget

2. Table orders

Order_id	Customer_name	Amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

SQL

```
SELECT customer_name,
       amount,
CASE
  WHEN amount >= 1000 THEN 'High Value'
  WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium_value'
  ELSE 'Low value'
END AS order_value_category
```

Customer_name	Amount	order_value_category
Alice	150	Low value
Bob	560	Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200	High value

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3. Table : employees

emp-id	emp-name	department	salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

SQL

```
SELECT emp_name,
       department,
       salary,
CASE
WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'
WHEN department = 'HR AND salary > 55000 THEN ' Experienced HR'
     ELSE 'Staff'
END AS position_level
FROM employees;
```

emp-name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

4. Table : Students

Student_id	Student_name	Score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

Student_name	Score	Grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

outcome Table

SQL

```
SELECT Student_name,
       Score
CASE
 WHEN score >= 90 THEN 'A'
 WHEN score BETWEEN 80 AND 89 THEN 'B'
 WHEN score BETWEEN 70 AND 79 THEN 'C'
 WHEN Score BETWEEN 60 AND 69 THEN 'D'
  ELSE 'F'
 END AS Grade
 FROM Students;
```

5. Table: deliveries

delivery_id	Delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

SQL

```
SELECT delivery_id,
       delivery_time_minutes
CASE
WHEN delivery_time_minutes <= 30 THEN 'Fast'
WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
    Else 'Late'
END AS performance
From deliveries;
```

delivery_id	delivery_time_minutes	Performance
1	45	On time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

6. Table : Tickets

ticked_id	issue_type	Priority
1	Login issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

SQL

```
SELECT issue_type,
       Priority,
CASE
  WHEN priority = 3 THEN 'High'
  WHEN priority = 2 THEN 'Medium'
  WHEN priority = 1 THEN 'Low'
END AS priority_label
FROM tickets;
```

Issue type	Priority	Priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password Reset	1	Low

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7. Table : attendance

Student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

SQL

```
SELECT student_id,
(days_present *100 / total_days) AS attendance percentage,

CASE
WHEN (days_present *100.0 / total_days) >=90 THEN 'Excellent'
WHEN (days_present *100.0/ total_days) BETWEEN 75 AND 89 THEN 'Good'
    ELSE 'Needs_improvement'
END AS attendane_status
FROM attendance;
```

Student_id	attendance_percentage	attendance_status
1	90%	Excellent
2	60%	Needs Improvement
3	96%	Excellent
4	50%	Needs Improvement
5	100%	Excellent

8. Table products-inventory

Product-id	Stock_qty
1	5
2	0
3	25
4	10
5	3

SQL

```
SELECT product_id,
       Stock_qty,
CASE
WHEN stock_qty = 0 THEN 'out of stock'
WHEN stock_qty BETWEEN 1 AND 5 THEN 'Low Stock'
 ELSE 'IN stock'
END AS Stock-Status
FROM products_inventory
```

Product_id	Stock_qty	Stock_Status
1	5	In Stock
2	0	Out of stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

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9. Table : Classes

Class_id	Subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

SQL

```
SELECT subject,
        enrolled_students,
 CASE
WHEN enrolled_students >=25 THEN 'Large'
WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
  ELSE 'small'
 END AS class_size_category
FROM classes;
```

Subject	enrolled_students	class_size_category
Maths	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

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10. Table: payments

Payment_id	Amount	Payment_method
1	50,00	Card
2	200,00	Cash
3	150,00	Card
4	75,00	Paypal
5	300,00	Cash

SQL

```
SELECT payment_id,
       payment_method,
       amount,
CASE
WHEN payment_method = 'cash' AND amount >=100 200
  THEN 'Eligible for discount'
~~END AS discount~~
  ELSE 'Not eligible'
END AS discount_eligibility
FROM payment;
```

Payment_id	Payment_method	Amount	discount_eligibility
1	Card	50	Not eligible
2	Cash	200	Eligible for discount
3	Card	150	Not eligible
4	Paypal	75	Not eligible
5	Cash	300	Eligible for discount

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